

COSC 363: Computer Graphics

Assignment 2

Max. Marks: 20

Due: 11:59pm, 25 May 2026

Aim

In this assignment you will implement a ray tracer that can handle different types of geometric objects and global illumination features, and demonstrate its capability in enhancing the visual realism of a rendered scene.

1. The Basic Ray Tracer (Max. marks: 8)

In labs 7 and 8, you will develop a simple ray tracer that can handle scenes containing planes and spheres. You will also implement methods for generating shadows and reflections. This assignment builds upon that ray tracer. As a minimum, your ray tracer should include the following features/objects.

- (a) A set of scene objects in a good spatial arrangement inside a fully enclosed environment. The scene enclosure must either be defined as:
 - six differently coloured or patterned axis-aligned planes which form a box around the scene, OR
 - a textured sky box/sphere. Lighting calculations must be disabled for the sky box/sphere.
- (b) At least one transparent object such as a sphere. (Note: A transparent object must not be modelled as a special case of a refractive surface with $\eta_1 = \eta_2 = 1$).
- (c) Objects must cast shadows. Transparent and refractive objects must have lighter shadows (implemented as the ambient lighting component plus a proportion of the diffuse and specular lighting components).
- (d) The scene must include at least one planar object containing mirror-like reflections of surrounding objects.
- (e) At least one surface in the scene must have a chequered pattern OR a texture applied.
 - If implementing a chequered pattern, the method used for generating a stripe pattern in Lab-8 may be extended for this.
 - If applying a texture, a different texture than the one from Lab-8 must be used, and the texture must map to a different region of the scene.

OpenGL functions should be used only for the final rendering of the ray traced image.

2. Extensions (Max. marks: 9)

This assignment will be marked out of 20, with 17 marks for the ray tracer and 3 marks for a report (see below). With just the basic features listed in Section 1 and a report, your ray tracer would earn at most 11 marks out of 20. To get more than that, you will need to implement additional features. The list of possible features and the

marks they would each gain if implemented correctly (up to a maximum of 9 total marks for extra features) are given below.

- Additional object types:
 - Single/double/truncated cone: 1 mark
 - Cylinder with clearly visible cap: 1 mark
 - Torus: 3 marks
- Refraction of light through an object (e.g. a refractive sphere): 1 mark
- Object-space transformations: 1.5 marks. The transformation(s) must change the object in some way that cannot be achieved by changing the object parameters (e.g. rotating a cylinder or cone off the y axis, or flattening a sphere)
- Anti-aliasing: 1 mark. The report must contain outputs of the ray tracer with and without anti-aliasing.
- Textured non-planar (curved) object: 1 mark.
- Spotlight: 0.5 marks.
- Stochastic sampling (soft shadows, depth of field, or rough surface reflections): 1 mark. The report must contain outputs of the ray tracer with and without stochastic sampling.
- Multiple light sources, including multiple shadows and multiple specular highlights generated by them: 1 mark
- Constructive Solid Geometry (CSG): up to 4 marks. The construction must use intersection or subtraction (not only union) and have correct normals. 2 marks for CSG combining only 2 objects, 3-4 marks for CSG combining 3+ objects. The combination of objects used must be specified in the report.

The marks associated with each feature should be taken to be indicative of the time and/or effort required to implement that a feature. The 8 marks for the minimum requirements are relatively easy to get when compared to the marks gained for some of the extension features. Marks will be given to only those features that are clearly visible in the ray traced output, and marks may be subtracted for any visual artefacts/rendering issues present. Each object must have correct normals and lighting.

3. Report (Max. marks: 3, No. of pages: 2 – 4)

The report should describe your ray tracer, including both its successes and its failures. It should include at least one image showing the ray tracer's capabilities. You may include additional images demonstrating various features of the ray traced scene. The report should clearly describe all extra features you have implemented in the ray tracer. For example, if you have implemented anti-aliasing in your ray tracer, you should describe its purpose, and how it is implemented.

You must also give an estimate of the time taken by your program to generate the output on your personal computer or lab machine. To keep run times to a minimum, you should do most of your development with small images (approx. 600x600 pixels), and low levels of recursion.

For this assignment, **you must not use any online/external code or libraries** except for `stb_image.h` for loading images. You may use any code provided within COSC363, and short (1-3 line) snippets provided by generative AI. Use of any data or

images from external resources must be acknowledged in the report. If you used generative AI, you must include a statement of acknowledgement (see section 4). Please include details of the build process/environment.

The report must contain the following declaration. **Assignment submissions without the declaration will not be accepted.**

Declaration

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

Name

Student ID

Date

4. Generative AI Usage

Generative AI Tools Are Permitted for Certain Parts of This Assessment



In this assessment, you are permitted to use generative artificial intelligence (AI) solely for the purposes of explaining the usage of computer graphics functions, answering questions about debugging, generating short code segments for reuse (1-3 lines), and grammar checking. No other use of generative AI is permitted. To assist with maintaining academic integrity, you must appropriately acknowledge any use of generative AI in your work. Please include a statement of acknowledgement with your work, clearly indicating which AI tools were used and how they contributed to your assessment.

5. Assignment Submission

Submit electronically (using *Learn*), the source code and any other supplementary files needed to run the program. Please also submit your report in PDF format. The files may all be packaged together and submitted as a single .zip file.

6. Miscellaneous

1. Check regularly on the *Learn* system forums for spec updates and clarifications.
2. This is not a group project. Each student must individually develop the program. In particular, students are not permitted to share program source code in any way. Standard departmental regulations regarding dishonest practices apply.